

Review list 1 - 170S - Spring 2019

Some of the key definitions and ideas that you are expected to know:

- 7.4 accuracy and [length of the confidence interval](#), their connection to the [sample size](#)
- 7.1, 7.2, 7.3 **definitions:** [one and two-sided confidence intervals](#) with [confidence level](#) $x\%$ (for an estimate $\hat{\theta}$ of a parameter θ); [confidence of a given interval](#)
- how to:** how to compute formulas for the confidence intervals for means, proportions, or based on definition; how to compare two sets of samples using confidence intervals
- 6.8 **definitions:** [prior, posterior distributions](#), [MAP estimator](#), [conditional expectation estimator](#)
- how to:** use Bayes formulas in the discrete and continuous cases; how to find and interpret posterior distribution, MAP, conditional expectation estimates
- 6.7 **definitions:** [sufficient statistics](#), densities in the exponential form
- how to:** use properties of sufficient statistics, find sufficient statistics from factorization theorem and using exponential form; check that a density have exponential form
- 6.5 **definitions:** [simple linear regression](#) problem (its formula and interpretation of each term), residuals
- how to:** find MLE for the coefficients of linear regression
- 6.4 **definitions:** [maximum likelihood function](#) and estimate; empirical moments; [method of moments](#) estimate; [biased and unbiased](#) estimates
- how to:** find maximum likelihood and method of moment estimators, how to check if an estimate is (un)biased
- 6.2, 6.3 **definitions:** [order statistics](#), [median](#), quantiles, quartiles, percentiles, sample quantiles (and formula for them), [quantile-quantile plot](#), box plot, interquartile range
- how to:** find the quantities and draw the plots mentioned above from given samples; how to find $P(X_{(i)} < 5)$ given density function
- 6.1 histograms and how to plot them, [sample mean](#) and [sample standard deviation](#)
- 5.5 (Only to the extent necessary for the chapter 6 and 7 material)
- concepts:** statements of the theorems 5.5-1, corollary 5.5-1, 5.5-2 and 5.5-3 and how to use them (for example, to find confidence intervals)
- Chi-square distribution as a distribution of the square of gaussian, distribution of the sum of two independent chi-square distributed random variables (without proofs). Properties of the normal distribution, central limit theorem, properties of expectations and variances. *See also 170E review list in Homework 1.*